PROPERNESS AND UNIVERSAL CLOSEDNESS ARE PRESERVED BY BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-morphisms/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot. Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Editorial note. The opening clause 'This is true by definition for universally closed morphisms' is a definitional check inside a three-part verification of a definition (universally closed, separated, finite type), not a deferral: each part is named and its source given, and properness is their conjunction.

Allowed standard prerequisites (editorial).

- Universal closedness is preserved by base change, by definition. the definition of a universally closed morphism, invoked at proof line 9924. Gives the first constituent directly.
- Separatedness is preserved by base change. Schemes, Lemma \ref{schemes-lemma-separated-permanence}, cited at proof lines 9925-9926. Gives the second constituent.
- Being of finite type is preserved by base change. Lemma \ref{lemma-base-change-finite-type}, cited at proof lines 9927-9928. Gives the third constituent.

Source reference. Original lemma 01W4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2469–2469. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01KU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2063–2063. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Date: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 01T4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2361–2361. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *The base change of a proper morphism is proper. The same is true for universally closed morphisms.*

Proof. This is true by definition for universally closed morphisms. It is true for separated morphisms (Schemes, Lemma 01KU). It is true for morphisms of finite type (Lemma 01T4). Hence it is true for proper morphisms. □